

MICHAEL LEVESQUE

SENIOR SOFTWARE ENGINEER

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Summary

Senior Software Engineer with 15+ years of experience building and optimizing software across macOS, Windows, web, and mobile platforms. Deep expertise in C++, cross-platform development, systems engineering, and performance optimization, with experience delivering software used by millions of users at Microsoft and Amazon. Proven track record of leading technical initiatives, improving system performance, and collaborating across engineering, product, and customer teams. Experienced in leveraging AI-assisted and agentic coding workflows to accelerate development, debugging, benchmarking, and technical analysis.

Experience

SOFTWARE ENGINEER II

MICROSOFT • Redmond, WA (2020 – 2026)

[Microsoft Teams](#) – C++, Windows, macOS, Direct3D, Metal, GTest, Instruments, Copilot

- Led the redesign of the Microsoft Teams render pipeline, working with the team to break a large monolithic compositor into smaller components with clearly defined responsibilities. Wrote the design document, helped divide the implementation work, reviewed changes, and supported testing and rollout.
- Designed the C++ architecture so common rendering logic could be shared across platforms, while Windows and macOS implementations handled Direct3D and Metal-specific behavior.
- Designed and implemented render pipeline optimizations that allowed composition buffers to retain existing content between render cycles, eliminating unnecessary background clears and enabling per-buffer frame tracking to reduce redundant draw calls by 8% for video meetings and 34% for video-based screenshare meetings.
- Expanded a cross-platform Teams performance benchmarking framework using agentic coding, adding automated log collection, decoding, and parsing for Windows and macOS, and visualizing performance metrics using interactive HTML reports with zoomable graphs and synchronized timeline markers.
- Integrated automated Instruments profiling and on-demand process sampling on macOS, enabling benchmarks to automatically attach to Teams, capture profiling traces, and collect diagnostic samples for investigating hangs and performance issues.

INDEPENDENT SOFTWARE DEVELOPMENT

Redmond, WA (2018 - 2020)

[3D/VFX Tools](#) • iOS Development, Xcode, Swift, JSON

- Designed and developed an iOS application using Swift and ARKit to capture physical camera-rig position and orientation with timestamps for use in visual-effects workflows.
- Built a synchronized recording workflow that captured audio alongside tracking data to assist with aligning camera movement to filmed scenes during post-production.
- Developed a recording UI for organizing captures by project, scene, and take, displaying live position and orientation data, and automatically organizing recorded files.
- Created 3ds Max/MaxScript tools to automate 3D modeling and animation workflows, including procedural generation of physics-enabled objects from splines to simulate loose copper wires that would have some sway but maintain their spline shape.

Experience

SOFTWARE DEVELOPMENT ENGINEER II

AMAZON • *Seattle, WA (2017 – 2018)*

[Seller Central: Manage Your Orders](#) – TypeScript, CSS, React, Redux

- Developed features for Amazon Seller Central's order-management experience, improving how sellers worldwide organize and manage customer orders.
- Built single-page application features using React and Redux for Seller Central experiences used by millions of sellers.
- Diagnosed and responded to high-severity production issues, performing root-cause investigation and deploying hotfixes.

GAME DEVELOPER

GSN GAMES • *San Francisco, CA (2014 – 2017)*

[Wheel of Fortune Slots: The Ultimate Collection](#) – Unity 3D, C#, JSON

- Designed a game-state architecture supporting parallel state machines to coordinate core game flow and user-experience behaviors.
- Designed and developed a configurable progression system for long-running GSN web games to improve player engagement by 10% and lifted monthly revenue by 20%.
- Designed and implemented a choreography system, enabling tech artists to build complex series of synchronized animations with animation triggers.
- Created a bonus wheel spin mechanic for players to earn rewards through a Wheel of Fortune branded wheel using positional equations and simulated spring physics.

SENIOR SOFTWARE ENGINEER

ZYNGA • *San Francisco, CA (2011 – 2014)*

[ChefVille](#) – Flash, Actionscript, PHP, JSON

- Led development of core game systems, such as Café Design Mode, allowing millions of players to customize their café layouts.
- Implemented grid-based collisions and specialized game object placement rules for game logic.
- Improved ChefVille content creation tools to increase productivity for product managers and designers.
- Led the full-stack development of key game features such as NPC behavior, game object placement and removal, cooking station upgrades, and pathfinding.

Skills

Languages: C++, C#, TypeScript, Swift

Platforms & Technologies: macOS, Windows, iOS, Direct3D, Metal, Instruments, React, Redux, GTest, Unity, ARKit

Engineering: Cross-Platform Development, Systems Engineering, Performance Optimization, Rendering, Computer Graphics, Profiling & Benchmarking, AI-Assisted Development / Agentic Coding

Education

Bachelor of Science in Computer Science

DigiPen Institute of Technology • *Redmond, WA (2005 – 2010)*